TELIPORT
Script Your Future
Season 4

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Case Study No. 3 – Adaptive QoS Engine (PulseQoS)

Team Details

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THINK. BUILD. CREATE. WIN

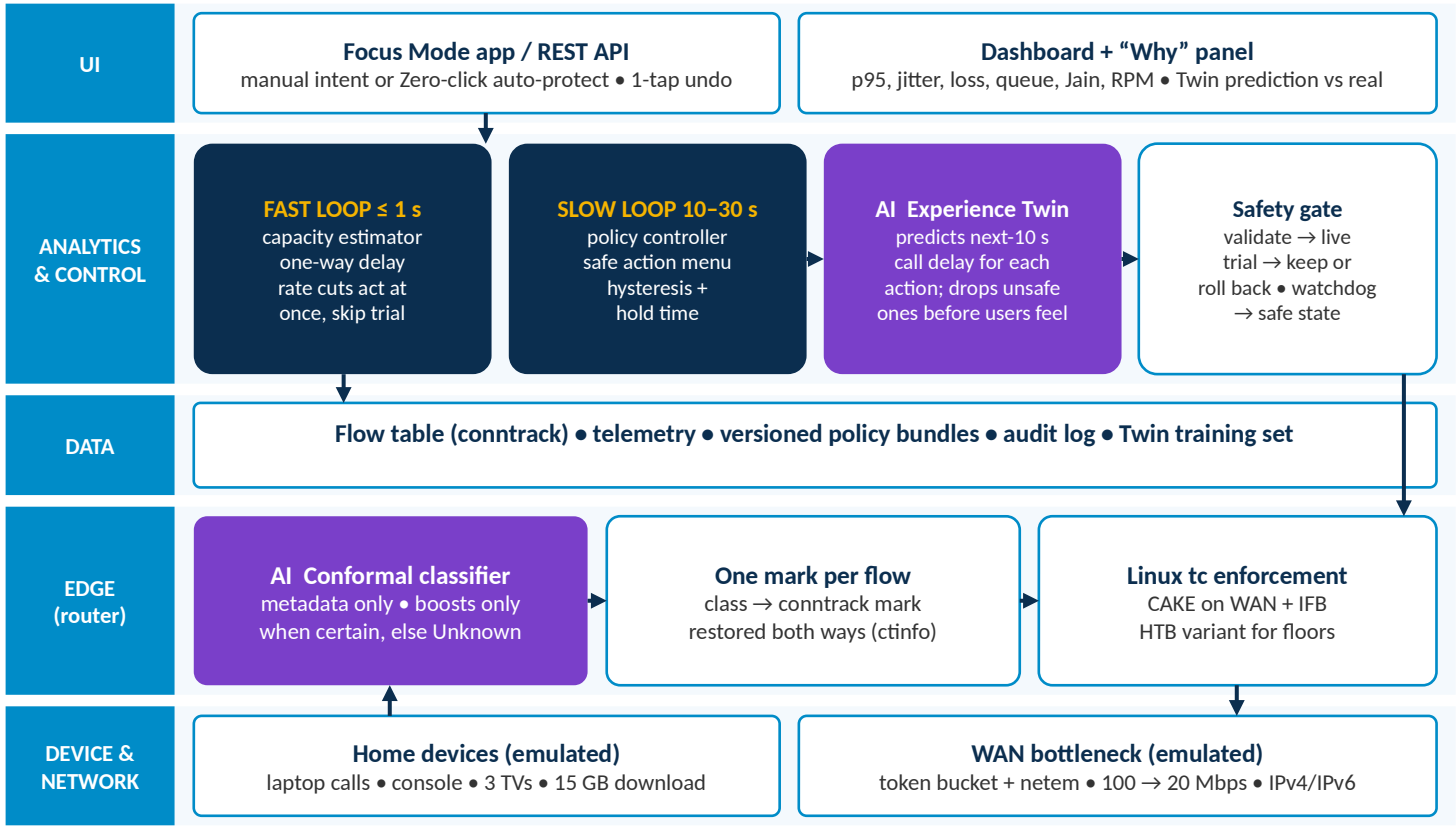
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PulseQoS: the call never notices the download

Intent-aware, self-verifying QoS controller for home broadband — payload-blind, built on open Linux tc

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ARCHITECTURE — one closed loop, two speeds



DESIGN RULES THAT KEEP IT CORRECT

- **Tins:** call audio + gaming → Voice; call video → Video (Voice tin is only 25%; RFC 8837); Unknown never boosted
- **Trust:** Internet DSCP is cleared on arrival; only our own conntrack mark sets priority
- **Link realism:** overhead compensation, upload ACK filter, per-direction one-way delay
- **Privacy & abuse:** no payload or domain logging; LAN-only authenticated intent API with budgets

THE AI ONLY PULSEQOS HAS — each must beat a rule-based twin

1 Conformal classifier. Boosts a flow only when its prediction set is a single call/game class, so wrong boosts stay ≤ a chosen rate (e.g. 1%) by construction on calibration-like traffic. **vs port rules**

2 Experience Twin. Learned model of the home link (trained offline in the lab, never by exploring on users) predicts each change's effect and rejects harmful ones before they go live. **metric: harm-seconds avoided**

3 Zero-click Focus. Detects a call starting on a busy link and protects it within seconds, charged to a Fairness Budget, with undo. **metric: time-to-protection**

No chatbot: the case asks AI to improve a measurable outcome, not add a chat interface. An AI part that loses its A/B test is switched off.

PILOT EVIDENCE — measured 27 Oct, not assumed

p95 call-probe delay during a 4-stream download (ms)



Download goodput identical: 18.7 of 20 Mbps — call protected, bulk not starved.
5 paired runs, Linux namespaces, 20 Mbit link; emulation without WAN delay, ideal classification. Round 2: CAKE, netem, 8 systems × 14 scenarios × 10 runs, incl. CAKE's own autorate.

Boundary: reference implementation in emulation, labelled as such. A product also needs router hardware, flow-offload handling, gigabit CPU budget, Wi-Fi airtime fairness and certification.